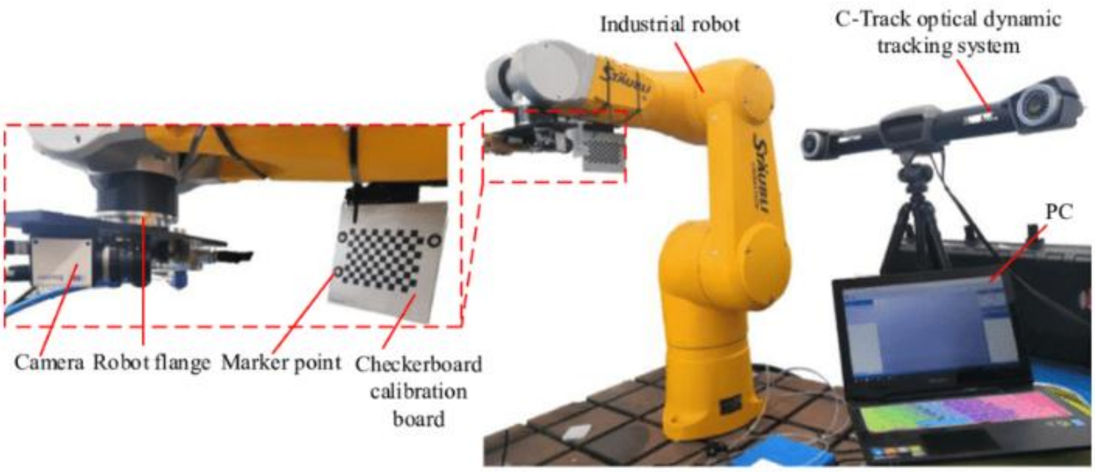
2. Checkboard-Free Hand–Eye Calibration

Hand–eye calibration is a fundamental problem in robotics and 3D vision that aims to determine the precise geometric relationship between a robot's end-effector ("hand") and a camera ("eye") attached to it. This transformation, typically represented as a rigid 6-DoF pose, is essential for tasks such as visual servoing, 3D reconstruction, object grasping, and human–robot interaction.

In practice, the calibration problem can be expressed as solving the equation **AX = XB**, where

- **A** represents the robot's motion between two configurations (in the robot base frame),
- **B** represents the corresponding camera motion (in the camera frame), and
- **X** is the unknown homogeneous transformation between the hand and eye.

Despite its apparent simplicity, real-world calibration is complicated by sensor noise, imperfect motion estimation, time synchronization errors, and non-linear optimization challenges. Accurate hand–eye calibration is therefore a critical prerequisite for high-precision robotic vision applications.



Traditionally, this calibration is performed using a known calibration target, such as a checkerboard. While effective, this approach can be error-prone in real-world scenarios, e.g., when the checkboard is not fully observed or even not available. Checkerboard-free methods overcome this limitation by leveraging natural features present in the environment to estimate the camera's motion, enabling more flexible and autonomous calibration. Nowadays, data-driven foundation models like DUSTt3R and VGGT can jointly estimate point maps and camera transformation, which make checkerboard-free methods feasible.

Basic

- Understand classical methods such as **Tsai–Lenz algorithm (1989)** and **Park–Martin dual quaternion method (1994)**. Implement a hand–eye calibration algorithm that estimates the transformation X from a set of paired motions (A_i, B_i) while B_i is ambiguous at scale (estimated from foundation models or

COLMAP). You can start with data provided by https://github.com/tomtang502/arm_3d_reconstruction.

- Combine estimating camera motion and hand-eye calibration algorithm and evaluate on provided synthetic data. The metrics are rotation error (in degrees) and translation error (in centimeters).
- Visualize the result by showing the transformed camera coordinate system relative to the robot arm in 3D (e.g., Open3D, pytransform3d, or matplotlib 3D).

Advanced Options

- **Noise-Robust Estimation:** Introduce realistic noise in robot and camera motions and design a robust optimization method (e.g., RANSAC, M-estimator) to handle outliers.
- **Deep Learning–Based Approach:** Implement or fine-tune a neural network that directly predicts the hand–eye transformation from paired motion trajectories or RGB-D frames.

Reference

1. (IEEE T-RA 1989) Tsai & Lenz: A new technique for fully autonomous and efficient 3D robotics hand–eye calibration
2. (IEEE T-RA 1994) Park & Martin: Robot sensor calibration: Solving AX = XB on the Euclidean group
3. (IROS 2017) Simultaneous hand-eye calibration and reconstruction
4. (RA-L 2024) Unifying Representation and Calibration With 3D Foundation Models
5. https://sites.google.com/andrew.cmu.edu/deep-hand-eye-calibration